

apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack■tip shielding effect under cyclic loading

Executive Technology Brief

Domain: mechanical_structural

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'fatigue fracture' concerning railway rail. User-submitted engineering problem (session input): "Why do rails fracture in service under fatigue loading?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2020). Illustrative records: 'How we can control the crack to propagate along the specified path feasibly?'; 'Steady-state propagation of a Mode III crack in couple stress elastic materials'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: crack tip shielding effect that correlates crack growth rate with the loading force range ΔF in addition to the stress■intensity range ΔK

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack■tip shielding effect under cyclic loading

Governing Model: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: APPLICABLE; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value – geometry factors must be applied from the actual design. Engaged variables: σ .

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation – use their measured plateau relations instead. Engaged variables: σ .

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.